TATA CONSULTANCY SERVICES
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CERSAI 2.0 – API Specifications

Document-Dedup

TATA Consultancy Services Ltd.

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ver. 1.2

Suraj Prajapati

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Contents

Copyright....2

Tata Code of Conduct 2

Document Revision History 3

List of Abbreviation 6

1. **Introduction** 7

2. Pre-requisites 8

3. Process Flow 9

Consuming the APIs 10

4. Steps for Registration for API 11

5. Registration of Security Interest API-Dedup 13

URL 13

API Request (Encrypted) 13

API Request (Un-encrypted) 14

API Request Attributes 15

API Response (Encrypted) 16

API Response (Un-encrypted) 17

API Response Attributes 17

API Server Side Validation Error Response (Encrypted) 18

API API Server Side Validation Error Response (Un-encrypted) 18

API Response Attributes 18

Annexure I: Sample Requests and Responses 19

Annexure II: Exception – Error Code and Description for Security Interest API DeDup 23

List of Tables

Table 1: List of Abbreviations 6

Table 2: De-Dup API Request Attributes 15

Table 3: De-Dup API Response Attributes 17

Table 4: Server Side Validation Error API Response Attributes 18

Table 5: Error Code and Description for Security Interest API DeDup 23

List of Abbreviation

This section provides the details of the abbreviations used in the document.

Table 1: List of Abbreviations

Abbreviation	Expanded Form
CERSAI	Central Registry of Securitisation Asset Reconstruction and Security Interest of India
API	Application Programming Interface
TCS	Tata Consultancy Services Limited
MD	Managing Director
SARFAESI	Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act, 2002
PUA	Primary User Administrator
RSA	Rivest-Shamir-Adleman Assymetric encryption algorithm
REST	Reprtesentational State Transfer
JSON	Javascript Object Notation
AES	Advanced Encryption Standard symmetric encryption algorithm
URL	Uniform Resource Locator
PROD	Production Environment
UAT	User Acceptance Testing Environment

1. Introduction

Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI) is a Government of India company licensed under section 8 of the Companies Act, 2013 with Govt. of India having a shareholding of 51% by the Central Government and select Public Sector Banks and the National Housing Bank also being shareholders of the Company. The objective of the company is to maintain and operate a Registration System for the purpose of registration of transactions of securitization, asset reconstruction of financial assets and creation of security interest over property, as contemplated under Chapter IV of the Securitization and Reconstruction of Financial Assets and Enforcement of Security Interest (SARFAESI) Act, 2002.

To provide this mechanism, a web-based system was developed for registration of transaction by financial institutions. It enables lenders and other stake holders to get real time current information regarding the collateral being mortgaged by borrower which helps in preventing fraudulent transactions arising out of same asset being mortgaged with multiple lenders. It also provides potential buyers of assets with information about any encumbrances on the assets.

The functionalities and the end users of the Central Registry have increased steadily over a period of time since the Central Registry was operationalised on 31st Mar 2011. CERSAI has developed certain APIs to cater to the increased load on the Central Registry Portal as well as to provide a better end-user experience. This document provides the specifications for the Security Interest registration API.

2. Pre-requisites

In order to consume the APIs exposed by CERSAI, the following pre-requisites must be fulfilled:

1. Consumer entity must be a registered entity in the Central Registry portal.
2. Consumer entity must have a valid class III digital signature.
3. Consumer entity will have to fill and submit the API Requisition Form.
4. Consumer entity must incorporate all the relevant master data required for consuming the APIs. The master data can be downloaded from the 'Downloads' section of the CERSAI portal.

3. Process Flow

The process flow for consuming CERSAI 2.0 APIs is mentioned below:

1. Any entity that wants to consume the CERSAI APIs, will have to first fill an API registration form through the portal. The PUA users of the entity will have the authorisation for filling this form.
2. A valid class III digital signature is required for performing this step. This digital signature will be required for signing all the API requests and hence, need to be plugged in / present on the consuming server all the time.
3. After filling in the necessary details and selecting the digital signature, the PUA will submit the form.
4. After CERSAI admin user has successfully approved the request, the system will generate:
    - 4.1. A new channel key. This channel key will be required to be passed in all API requests.
    - 4.2. A new user Id and password, to be used exclusively for consuming the APIs.
    - 4.3. A new RSA key pair. The public key will be shared with the institution while the private key will be maintained in the system.

Consuming the APIs

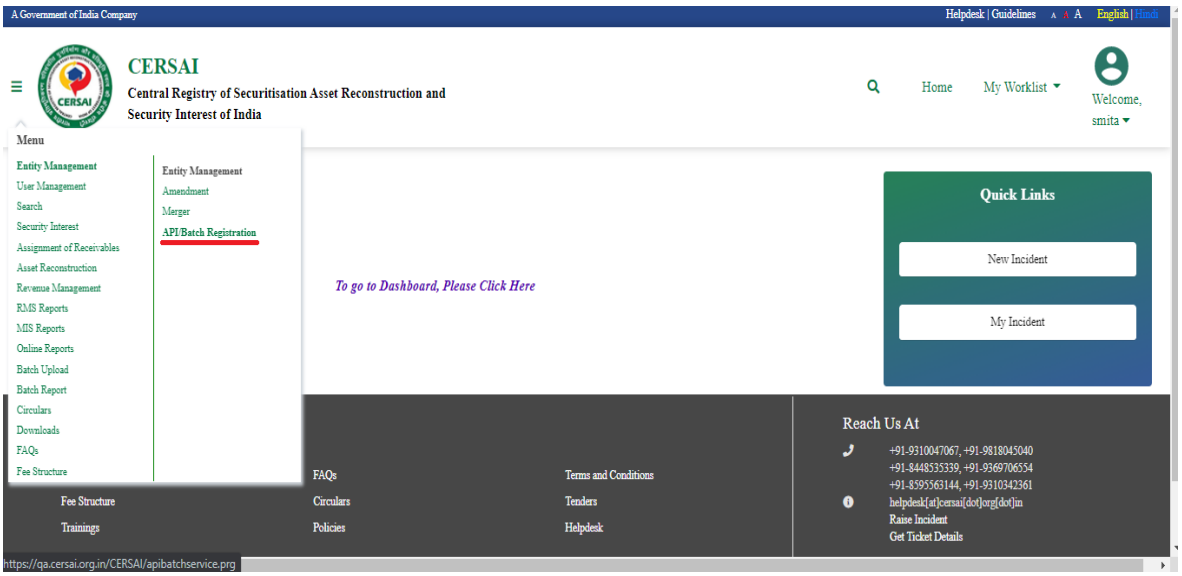
1. For consuming any API, the consumer entity will have to first test the API in the non-production environment of CERSAI. Only after successfully testing the API in the non-production environment, the production environment will be enabled for the consumer entity.
2. All the APIs exposed by CERSAI will be REST APIs. The request and response payloads will be in JSON format.
3. Each API payload will be having two sections – 'header' and 'body'.
4. The header section will contain the attributes related to authentication and authorisation of the API request.
5. The body section will further contain three main attributes – 'additional info' ,'session key' and 'request payload'.
6. For consuming the APIs, first of all, the consuming system will have to digitally sign the DeDupSIRegistrationRequest Object using the digital signature that was registered on the CERSAI portal while filling the API registration form and then pass the signed value in RequestHash attribute of RequestPayload.
7. After DeDupSIRegistrationRequest is successfully digitally signed and RequestHash is created, a new unique AES256 session key and 16-byte IV parameter will have to be generated. The RequestPayload (containing DeDupSIRegistrationRequest and Request hash) will then have to be encrypted using this session key and IV parameter.
8. The session key, IV parameter and password will have to be then encrypted using the public key shared with the entity during the API registration process.
9. Encode the encrypted data to base-64 and invoke the API.
10. The response from CERSAI will only be encrypted using session key received in the API Request and encoded to base-64.
11. This API will be consumed after the below mentioned conditions:
    - 11.1. You must have integrated with CERSAI for Registration of Security Interest API.
    - 11.2. If in Registration of Security Interest API, you have passed **isregistrationDedupBypassed** as **"False"** or **"null"**
    - 11.3. You will receive similar assets in response from Registration of Security Interest API.

4. Steps for Registration for API

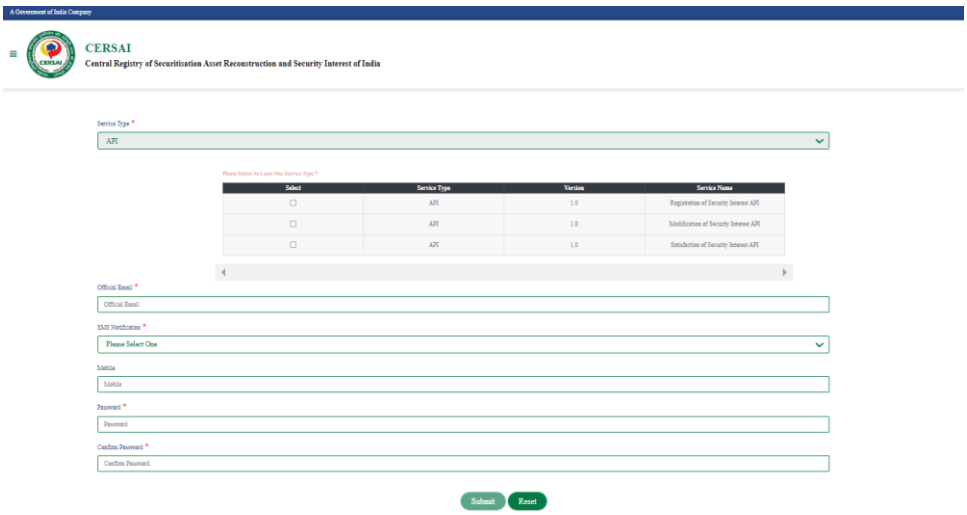
4.1. Open URL:- https://www.qa.cersai.org.in/CERSAI/home.prg

4.2. Login using User ID and Password of Primary User Admin.

4.3. Once login, Under Menu option, Choose API/Batch registration under Entity Management Option.



4.4 Form will open, Fill the required details (Password entered on the screen will be of API user and will be used for every API request)



4.5 Click on Submit button once data is entered.

4.6 Pop-up for digital signature will appear where the user has to use different digital signature for the registration process as this Digital Signature will be linked to the API user and will be used in every request to digitally sign the payload of every request.

4.7 Once above steps will be done. You will get a system generated mail from our side, which include a Channel Key, RSA Public Key and API User ID.

5. Registration of Security Interest API-Dedup

This section describes the API for registration of Security Interest / Attachement order transactions.

URL

The URL for the API will be shared separately with the approved entities.

API Request (Encrypted)

```
{
      "header": {
        "authinfo": {
        "LoginParams": [
            "APILOGIN",
                "<<API User Id>>",
                "<<API User Password, create hash using SHA256, encrypt using pre-shared RSA public key, base-64 encoded>>"
        ],
        "UserType": "APIUSER",
        "AuthenticationType": "AUTH_API"
        }
    },
    "body": {
        "AdditionalInfo": {
            "EntityCode": "<<Consumer entity code, plain text>>",
            "IV": "<<Create 16-byte IV parameter, Convert to String, encrypt using pre-shared RSA public key, base-64 encoded>>"

        },
        "SessionKey": "<<Unique AES256 key, encryptusing pre-shared RSA public key, base-64 encoded >>",
        "RequestPayload": "<<encrypt using AES256 session key and 16-byte IV parameter, base-64 encoded>>"
    }
}
```

API Request (Un-encrypted)


```
{
"header": {
    "authinfo": {
        "LoginParams": [
            "APILOGIN",
            "<<API User Id>>",
            "<<API User Password, encrypted using pre- shared RSA public key
            and base-64 encoded>>"
        ],
        "UserType": "APIUSER",
        "AuthenticationType": "AUTH_API"
    }
},
"body": {
    "AdditionalInfo": {
        "EntityCode": "<<Consumer entity code, plain text>>",
        "IV": "<<Create 16-byte IV parameter, Convert to String,
        encrypt using pre-shared RSA public key, base-64 encoded>>"
    },
    "SessionKey": "<<Unique AES256 key, encrypted using pre-shared RSA
    public key and base-64 encoded>>",
    "RequestPayload": {
        "DeDupSIRegistrationRequest": {
            "selectedAssetId": "2XXXXXXXXXX4",
            "deDupStatus": "S",
            "pgResponse": "",
            "userRemarks": "",
            "transaction": {
                "transactionId": "2XXXXXXXXXXX7"
            }
        },
        "RequestHash": "MIIGbwYJKoZIhvcNAQcCoIIGYDCCBlwCAQExCzAJBg /…/ QlmUoW9+f/
        +JVQkw=="
    }
}
}
```

API Request Attributes

Table 2: De-Dup API Request Attributes

Sr. No.	Attribute Name	Data Type	Is Mandatory?	Remarks
	header			
	authinfo			
1	LoginParams			
1a	APILOGIN	Constant	Yes	Constant value: “APILOGIN”
1b	API User Id	String	Yes	API User Id
1c	API User Password	String	Yes	API Password
2	UserType	Constant	Yes	Constant value: “APIUSER”
3	AuthenticationType	Constant	Yes	Constant value: “AUTH_API”
	body			
4	**AdditionalInfo**			
4a	EntityCode	String	Yes	CERSAI entity code
4b	IV	String	Yes	16-byte IV parameter
5	SessionKey	String	Yes	Unique AES256 key
	RequestPayload			
6	**DeDupSIregistrationRequest**			
6a	selectedAssetId	Number	Yes	Mandatory in case of deDupStatus=S
6b	deDupStatus	String	Yes	Any one value among ‘S’ or ‘F’ where ‘S’ stands for Selected assest and‘F’ stands for Forceful insertion
6c	pgResponse	String	No	Will be ‘Null’ in case of API Request
6d	userRemarks	String	No	Will be ‘Null’ in case of API Request
	transaction			
7	transactionId	Number	Yes	Transaction ID which will be received in response from Registration of Security Interest API
8	**RequestHash**			
^: Values must be picked from the respective CERSAI masters.				

API Response (Encrypted)


```
{
    "header": {
        "transactionId": "91pfj01XXXXXXXXXXXX010",
        "status": "success"
    },
    "body": {
        "APIResponse":"<<encrypted using the AES256 session key and
        base-64 encoded>>"
    }
}
```


Note:

If users receive response in html encoded format for eg.

{"header":{"transactionid":"haf3l01666166440144133","status":"success"},"body":{"APIResponse":"HVB4a6lMVIIM83HzdvUfYEtyyTTXwkBOAJKQDgVNqP0Oe2ZULe+d3eb04ZJ44VJXTBFQjdnvMbJO2lmW4MMCmsRpDDghb2bltKhG7Jm5N7UikpcUgXGN2xMC/P29FBGYkxxy/o+53S5WbpMtxlFh1HDX7OttlyXVgTVwG5ryD0E3dkGHcCr0r2hp7efsfU6bFQti4/tTsTxDcFzG7uSG1WbV9zzMYj145TnFYI/kuIHgyWpBCtogquXvr3Yw0YWsLHvyGNTVGbpLnloJwtJ0b+oH9slsnCfN/khKff0DO8XmXzwetAhQX3hl8SsazqF9GohROM2TrfEwu0q4ErJY5FbAm9qpd6OoXRAjPQyYKHf27YUe7BjkpXqrYxzN6FsFE6SrWwE2zfJsnSQVSWmJLTHRIFWS+9gkwXjApp0gzhynk2L1EbowN+1Vh7C4t4/FnGh2oDohymPp/mqhXRZYlAeQ6Cmk4SLw4IPPXzH6QmpibvmIdi/FejTI6EJXlgSlMoEYbXxg6W7F7+c+KrejGuBZl8KeIm5Y6JT8kO9IPZ+k7XAzoGfIuEAs3T/gPeNZ8/OaaUM/s3OImE8hoVsEn5ktvJgLA8QCRghxTc9haaUHc9Y8vrcfMzJfj0mr2Ofl2EMzxtrtVwhsd5ZZutm8KaDyZlJjDykSPZdPfqF54rGLb5FoZ6HrdJNf4aUMqCLJG2FOE9N7nGWNvxTy/1lvNEw3xcwrcz//k22dQlSRRPydkClDQdQiAOX8Jb2+65qMaZAYtXIBrrYB726ft4mNkWukpwmOk8qEY/yG2TAAxVcIkIPTlw7FvCRxF+7fW8bhchGNH5/LgLsqFhYx7Axq/9PM69lRal+w24zP3q+2TWiBql8Er5jC9GOCNM7D6OU4OILnPruD8XXI9bGSuuZQfg=="}}

User needs to perform html decoding, if decrypted 'APIResponse' is received in above format.

Kindly include following line, in case user is using java:

StringEscapeUtils.*unescapeHtml*("<Decrypted response in HTML encoded format>");

API Response (Un-encrypted)


```
{
    "header": {
        "transactionId": "91pfj01XXXXXXXXXXXX010",
        "status": "success"
    },
    "body": {
        "APIResponse": {
            "transationStatus": "C",
            "responseMessage": "Successful SI Registration",
            "transactionId": 2001XXXXXX46,
            "assetId": 2005XXXXXX57,
            "securityInterestId": 4010XXXXXX32
        }
    }
}
```


API Response Attributes

Table 3: De-Dup API Response Attributes

Sr. No.	Attribute Name	Data Type	Is Mandatory?	Remarks
	header			
1	transactionid	String	Yes	CERSAI Request transaction Id
2	Status	String	Yes	Status of the API request
	body			
3	transationStatus	String	Yes	Status of the Security Interest transaction
4	responseMessage	String	Yes	SI Registration response message
5	transactionId	Number	Yes	Security Interest registration transaction Id
6	assetId	Number	Conditional	Asset Id in case of successful registration
7	securityInterestId	Number	Conditional	SI Id in case of successful registration

API Server Side Validation Error Response (Encrypted)


```
{
    "header": {
        "errorMessage": "Server Side Validation Error in
        SI Registration with Deduplication",
        "errorCode": "21012",
        "transactionid": "l6r9i01668423780877133",
        "status": "error",
        "errorDetails": "{\"Failure\":\"{\\\"APIResponse\\\":\\\"tT/Vig85qZJ77hdazCNZ
        gUADY+yXCqG1Ta6Iz1g37jYzU4KHKUtiaY2JIF6cAA6LlJJ2lHUGM3MCYV9TAnGYDA==\\\"}\"}"
    }
}
```


API API Server Side Validation Error Response (Un-encrypted)


```
{
   "header": {
        "errorMessage": "Server Side Validation Error in
        SI Registration with Deduplication ",
        "errorCode": "21012",
        "transactionid": "l6r9i01668423780877133",
        "status": "error",
        "errorDetails": "{\"Failure\":\"{\\\"APIResponse\\\"selectedAssetId": "The
        Value Cannot Be Null / Empty"\\\"}\"}"
    }
}
```


API Response Attributes

Table 4: Server Side Validation Error API Response Attributes

Sr. No.	Attribute Name	Data Type	Is Mandatory?	Remarks
	header			
1	errorMessage	String	Yes	CERSAI API Error Message
2	errorCode	String	Yes	Status of the API request (21001 For Server Side Validation Error in Security Interest Registration API)
3	transactionid	String	Yes	CERSAI Request transaction Id
4	Status	String	Yes	Status of the API request
5	errorDetails	String	Yes	Detailed Error Message.

Annexure I: Sample Requests and Responses

- De-Dup Status: **Selected**

Request:


```
{
    "header": {
        "authinfo": {
            "LoginParams": [
                "APILOGIN",
                "<<API User Id>>",
                "<<API User Password, create hash using SHA256, encrypt using
               pre-shared RSA public key, base-64 encoded>>"
            ],
            "UserType": "APIUSER",
            "AuthenticationType": "AUTH_API"
        }
    },
    "body": {
        "AdditionalInfo": {
            "EntityCode": "<<Consumer entity code, plain text>>",
            "IV": "<<Create 16-byte IV parameter, Convert to String, encrypt using
           pre-shared RSA public key, base-64 encoded>>"
        },
        "SessionKey": "<<Unique AES256 key, encrypt using pre-shared RSA public key,
       base-64 encoded >>",
       "RequestPayload": {
            "DeDupSIRegistrationRequest": {
                "selectedAssetId": "2XXXXXXXXXX4",
                "deDupStatus": "S",
                "pgResponse": "",
                "userRemarks": "",
                "transaction": {
                    "transactionId": "2XXXXXXXXXXX7"
                }
            },
            "RequestHash": "MIIGbwYJKoZIhvcNAQcCoIIGYDCCBlwCAQExCzAJBg /…/ QlmUoW9+f/
            +JVQkw=="
        }

    }
}
```

Response:


```
{
    "header": {
        "transactionid": "mm0n1XXXXXXXXXXXXXXX10",
        "status": "success"
    },
    "body": {
       "APIResponse": {
            "transationStatus": "C",
            "responseMessage": "Successful SI Registration",
            "transactionId": 2001XXXXXX46,
            "assetId": 2005XXXXXX57,
            "securityInterestId": 4010XXXXXX32
       }
    }

}
```

- De-Dup Status: **Forceful**

Request:


```
{
    "header": {
        "authinfo": {
            "LoginParams": [
                "APILOGIN",
                "<<API User Id>>",
                "<<API User Password, create hash using SHA256, encrypt using
               pre-shared RSA public key, base-64 encoded>>"
            ],
            "UserType": "APIUSER",
            "AuthenticationType": "AUTH_API"
        }
    },
    "body": {
        "AdditionalInfo": {
            "EntityCode": "<<Consumer entity code, plain text>>",
            "IV": "<<Create 16-byte IV parameter, Convert to String, encrypt using
           pre-shared RSA public key, base-64 encoded>>"
        },
        "SessionKey": "<<Unique AES256 key, encrypt using pre-shared RSA public key,
       base-64 encoded >>",
       "RequestPayload": {
            "DeDupSIRegistrationRequest": {
                "deDupStatus": "F",
                "pgResponse": "",
                "userRemarks": "",
                "transaction": {
                    "transactionId": "2XXXXXXXXXXX7"
                }
            },
            "RequestHash": "MIIGbwYJKoZIhvcNAQcCoIIGYDCCBlwCAQExCzAJBg /…/ QlmUoW9+f/
            +JVQkw=="
        }

    }
}
```

Response:


```
{
    "header": {
        "transactionid": "mm0n1XXXXXXXXXXXXXXX10",
        "status": "success"
    },
    "body": {
       "APIResponse": {
            "transationStatus": "C",
            "responseMessage": "Successful SI Registration",
            "transactionId": 2001XXXXXX46,
            "assetId": 2005XXXXXX57,
            "securityInterestId": 4010XXXXXX32
       }
    }

}
```

Annexure II: Exception – Error Code and Description **for Security Interest API DeDup**

Table 5: Error Code and Description for Security Interest API DeDup

SR NO	ERROR CODE	ERROR MESSAGE	DESCRIPTION
1	1002	Action ID is invalid	URL/ Service Name is incorrect.
2	15000	Internal Server Error Occurred. Please try after some time.	Exception occurred as Channel Key is invalid.
3	70008	Invalid Security Interest Registration De-Dup Request	Security Interest Registration De-Dup Request is invalid.
4	70007	Failure while processing API request	API Request is incorrect/ API Request structure is incorrect/ API Request is not encrypted properly.
5	8726	Unsuccessful Login! Please try to Login again.	API Login Id is invalid.
6	21012	Server Side Validation Error in SI Registration with Deduplication	Server side validation error. Request Json didn't pass required validations.
7	41901	Something went wrong while verifying your Digital Signature. Please try again.	No signature attached with the Requesthash
8	41902	Digital Signature Verification Failed.	Digital Signature hash verification failed while verifying Security Interest signed hash.
9	41903	Your Digital Signature has been revoked.	Your Digital Signature has been revoked.
10	41904	Your Digital Signature has expired.	User Digital Signature has expired.
11	70001	Failure while decrypting request payload	API Request payload is incorrect/ API Request payload is not encrypted properly.
12	21010	Entity has insufficient Balance	Entity has insufficient Balance (In case of UAT testing, kindly connect with Technical Team CERSAI).